

Julien Tierny

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PROFESSIONAL EXPERIENCE

- 2021 – Present ★ *CNRS Research Director (DR2)*
Topological Data Analysis and Visualization,
Sorbonne Université, LIP6, Paris, France.
- 2023 – Present ★ *Professor (part-time, PCC)*
Computer Science Department,
Ecole Polytechnique, Palaiseau, France.
- 2010 – 2021 ★ *CNRS Research Scientist (CR)*, Topological Data Analysis and Visualization,
2014-2021: Sorbonne Université, LIP6, Paris, France.
2010-2014: Telecom ParisTech, LTCI, Paris, France.
- 2008 – 2010 ★ *Post-doctoral research associate* (computational topology for visualization and graphics),
Scientific Computing and Imaging Institute, University of Utah, USA;

EDUCATION

- 2016 ★ *Habilitation in Computer Science*, Sorbonne Université, France;
Thesis title: "*Contributions to Topological Data Analysis for Scientific Visualization*";
Committee: I. Bloch, J.-D. Fekete, P. Frey, H. Hagen, C. Johnson, B. Lévy, P. Ricoux, W. Schroeder.
- 2005 – 2008 ★ *Ph.D. in Computer Science*, Lille University, France;
Thesis title: "*Reeb graph based 3D shape modeling and applications*";
Committee: A. Baskurt, B. Lévy, C. Labit, A. Srivastava, S. Tison.
Advisors: M. Daoudi and J.-P. Vandeberre.
Summa cum laude ("*Mention très honorable*").
- 2005 ★ *M. Sc. in Computer Science*, Lille University;
Summa cum laude ("*Mention très bien*"), valedictorian.
★ *Engineer degree at TELECOM Lille 1*;
(French "*Grande École*", Information Technology Institute, License and Master);
Specialization: Multimedia software engineering, valedictorian.
- 2000 ★ French scientific *Baccalauréat* ;
Summa cum laude ("*Mention très bien*").

AWARDS

- 2025 ★ **Best Paper Honorable Mention Award**, IEEE VIS 2025;
- 2023 ★ **French Academy of Sciences, Young Researcher Award** ("*Prix Espoir IMT - Académie des sciences*");
★ **Best Paper Honorable Mention Award**, IEEE VIS 2023;
- 2019 ★ **European Research Council (ERC) consolidator grant**;
★ **Best Paper Honorable Mention Award**, IEEE VIS 2019;
★ **Best Paper Award**, IEEE LDAV 2019;
- 2018 ★ **Best Paper Honorable Mention Award**, IEEE LDAV 2018;
- 2017 ★ **Best Paper Honorable Mention Award**, IEEE VIS 2017;
★ Selection of our open-source software platform **TTK** (Topology ToolKit) among O'Reilly's daily **Four short links**.
★ Selection of my paper "Jacobi Fiber Surfaces for Bivariate Reeb Space Computation" as a notable article in computing in 2016 by the journal **ACM Computing Reviews**.
★ Selection of my paper "Jacobi Fiber Surfaces for Bivariate Reeb Space Computation" as a notable highlight in visualization by the magazine **IEEE Computing Now** (February 2017 issue)
- 2016 ★ **Best Paper Award**, IEEE VIS 2016;
★ **Honorable Mention Award**, IEEE Scientific Visualization Contest 2016;
- 2013 ★ **Best Paper Award**, Eurographics Symposium on Parallel Graphics and Visualization 2013.
- 2008 ★ **Fulbright research fellowship** (US Department of State), Lavoisier research fellowship (France);
- 2005 ★ French Research Ministry Ph.D. fellowship;
★ **IBM Top Student Recognition Event** (EMEA), Stuttgart, Germany. Final contest winner (in group).

A W A R D S (continued)

2004 * French Research Ministry M.Sc. Excellence Fellowship.

R E S E A R C H

Research Interests

Research fields * Topological Data Analysis, Visualization, Ensemble Data;
Computational Topology, Morse Theory;

Funded Projects

ERC (Consolidator)	* <i>"In-situ Topological Reduction of Scientific 3D Data"</i> (TORI). Principal investigator, September 2020 - September 2026. http://erc-tori.github.io/
CNRS - University of Arizona ANR H2020-FET	* Joint Ph.D. fellowship, co-principal investigator, Late 2022 - Late 2025. * <i>"Tremplin"</i> , Principal investigator, Late 2019 - Early 2021. * <i>"Vestec: Visual Exploration and Sampling ToolKit for Extreme Computing"</i> Work package leader, March 2018 - March 2021.
CIFRE (Total)	* <i>"Topological Reduction for Very Large Data Analysis"</i> Principal investigator, July 2016 - July 2019.
CIFRE (Kitware)	* <i>"In-situ Topo"</i> : In-situ Topological Data Analysis. Principal investigator, February 2016 - February 2019.
FSN	* <i>"AVIDO"</i> : In-situ uncertain data analysis and visualization. Local investigator, October 2015 - October 2018.
ANR (CONTINT)	* <i>"CrABEx"</i> : Example-based 3D modeling support. Local investigator, November 2013 - November 2017.
RTRA-Digiteo	* <i>"Uncertain Topo-Vis"</i> : Visualization of Uncertain Scalar Fields through Topology Analysis. Principal investigator, May 2013 - Octobre 2014.
Ph.D. Students	* Théophane Loloum, October 2025 – October 2028. Co-advised with: Fabien Vivodtzev. * Lorenzo Brucato, October 2025 – October 2028. Co-advised with: Fateh Kaakai. * Mattéo Clémot, October 2023 – October 2026. Co-advised with: Julie Digne. * Sébastien Tchitchek, October 2023 – October 2026. * Mohamed Kissi, October 2022 – October 2025. * Keanu Sisouk, October 2022 – October 2025. Co-advised with: Julie Delon. * Eve Le Guillou, April 2022 – April 2025. Co-advised with: Pierre Fortin. * Mathieu Pont, October 2020 – October 2023. * Jules Vidal, September 2018 - September 2021. * Maxime Soler, July 2016 - July 2019. Co-advised with: Mélanie Plainchault and Bruno Conche. * Charles Gueunet, February 2016 - February 2019. Co-advised with: Pierre Fortin and Julien Jomier. * Ana-Maria Vintescu, May 2014 - May 2017. Co-advised with: Florent Dupont, Guillaume Lavoué, Pooran Memari. * Mariem Gargouri, November 2011 - June 2015. Co-advised with: Elsa Angelini, Erwan Jolivet, Philippe Petit. * Brian Summa, Doctoral Internship 2011, Ongoing collaborations.
Post-doctoral researchers	* Sylvain Gerbaud (October 2024 – Present). Co-advised with Pierre Fortin. * Mariam Pirashvili (October 2020 – October 2021, now Lecturer at the University of Plymouth). Co-advised with Steve Oudot. * Daisuke Sakurai (September 2016 - April 2017, now an associate professor at Kyushu University), Co-advised with Julie Delon. * David Guenther (May 2013 - October 2014, now a software engineer at Sirona Dentals Inc.).
Engineers	* Thomas Daniel (September 2024 – December 2025). Now a Ph.D. student in Stuttgart. * Alexandre Talon (2023 – 2024).

RESEARCH (continued)

- ★ Pierre Guilloux (February 2019 – January 2023).
Now a research engineer at Mines PSL.
- ★ Hugo Manet (October 2020 - 2021).
- ★ Guillaume Favelier (2016 - 2019, now a research engineer at INRIA).

Collaborations

International Collaborations

- ★ TU Kaiserslautern, University of Arizona, University of Warsaw, Tulane University, DLR, University of Utah, Clemson University, University of Leeds, Universidade de Sao Paulo, New York University, Lawrence Livermore National Laboratory.

National Collaborations

Visiting Professors

- ★ Kitware, Total, CEA, Renault.
- ★ Joshua Aaron Levine, University of Arizona (USA), July 2023;
- ★ Joshua Aaron Levine, University of Arizona (USA), June 2022;
- ★ Joshua Aaron Levine, University of Arizona (USA), September 2016;
- ★ Joshua Aaron Levine, Clemson University (USA), May-June 2013;
- ★ Valerio Pascucci, SCI Institute, University of Utah (USA), May-June 2011;

International Publications

Theses

- ★ **Julien Tierny**
“Contributions to Topological Data Analysis for Scientific Visualization”,
Habilitation thesis,
Committee: I. Bloch (President), J.-D. Fekete (Committee), P. Frey (Committee), H. Hagen (Reviewer), C. Johnson (Reviewer), B. Lévy (Reviewer), P. Ricoux (Committee), W. Schroeder (Committee).
Sorbonne Université, April 2016.
- ★ **Julien Tierny**
“Reeb graph based 3D shape modeling and applications”,
Ph.D. thesis,
Committee: S. Tison (President), A. Baskurt (Reviewer), B. Lévy (Reviewer), C. Labit (Committee), A. Srivastava (Committee), M. Daoudi (Advisor), J.P. Vandeboire (Co-advisor).
Lille University, October 2008.

Monograph

- ★ **Julien Tierny**
“Topological Data Analysis for Scientific Visualization”,
Springer (Mathematics and Visualization series), 2018.
ISBN 978-3-319-71506-3.

Edited books

- ★ Ingrid Hotz, Talha Bin Masood, Filip Sadlo, and **Julien Tierny**,
“Topological Methods in Data Analysis and Visualization VI”, Springer, 2021 (ISBN: 978-3-030-83499-9).
- ★ Valerio Pascucci, Xavier Tricoche, Hans Hagen, and **Julien Tierny**,
“Topological Methods in Data Analysis and Visualization: Theory, Algorithms and Applications”, Springer, 2011 (ISBN: 978-3-642-15013-5).

Journals

- ★ Théophane Loloum, Fabien Vivodtzev, David Hébert, Baptiste Reynier, Michel Arigoni, **Julien Tierny**
“DebrisTracer: Reliable Tracking in Hypervelocity Impact Fast Imaging”.
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2026.
- ★ Mattéo Clémot, Julie Digne, **Julien Tierny**
“Topological Autoencoders++: Fast and Accurate Cycle-Aware Dimensionality Reduction”.
IEEE Transactions on Visualization and Computer Graphics
Accepted, to be presented at IEEE 2026.
- ★ Keanu Sisouk, Eloi Tanguy, Julie Delon, **Julien Tierny**
“Robust Barycenters of Persistence Diagrams”.
IEEE Transactions on Visualization and Computer Graphics
Accepted, to be presented at IEEE 2026.
- ★ Michael Will, Jonas Lukasczyk, **Julien Tierny**, Attila Gyulassy, Gunther Weber, Hamish Carr, Christoph Garth
“Scalable Computation of Topological Abstractions for Scalar Data”.
Computer Graphics Forum
EuroVis STAR 2026.
- ★ Eve Le Guillou, Pierre Fortin, **Julien Tierny**,
“Distributed Discrete Morse Sandwich: Efficient Computation of Persistence Diagrams for Massive Scalar Data”.
IEEE Transactions on Parallel and Distributed Systems
Accepted, 2025.

RESEARCH (continued)

- ★ Thomas Daniel, Malgorzata Olejniczak **Julien Tierny**,
“*BondMatcher: H-Bond Stability Analysis in Molecular Systems*”.
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2025.
Best Paper Honorable Mention Award.
- ★ Alexandre Janin, Nicolas Coltice, Nicolas Chamot-Rooke, **Julien Tierny**
“*Geodynamics of a global plate reorganization from topological data analysis*”.
Nature Geoscience, 2025.
- ★ Keanu Sisouk, Julie Delon, **Julien Tierny**,
“*A User’s Guide to Sampling Strategies for Sliced Optimal Transport*”.
Transactions on Machine Learning Research, 2025.
- ★ Mohamed Kissi, Mathieu Pont, Joshua A. Levine, **Julien Tierny**,
“*A Practical Solver for Scalar Data Topological Simplification*”.
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2024.
- ★ Tanner Finken, **Julien Tierny**, Joshua A. Levine,
“*Localized Evaluation for Constructing Discrete Vector Fields*”.
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2024.
- ★ Eve Le Guillou, Michael Will, Pierre Guillou, Jonas Lukasczyk, Pierre Fortin,
Christoph Garth, **Julien Tierny**,
“*TTK is Getting MPI-Ready*”.
IEEE Transactions on Visualization and Computer Graphics
Accepted, to be presented at IEEE VIS 2024.
- ★ Mathieu Pont, **Julien Tierny**,
“*Wasserstein Auto-Encoders of Merge Trees (and Persistence Diagrams)*”.
IEEE Transactions on Visualization and Computer Graphics
Accepted, to be presented at IEEE VIS 2024.
- ★ Keanu Sisouk, Julie Delon, **Julien Tierny**,
“*Wasserstein Dictionaries of Persistence Diagrams*”.
IEEE Transactions on Visualization and Computer Graphics
Accepted, to be presented at IEEE VIS 2024.
- ★ Markus Flatken, Artur Podobas, Riccardo Fellegara, Achim Basermann, Johannes
Holke, David Knapp, Max Kontak, Christian Krullikowski, Michael Nolde, Nick
Brown, Rupert Nash, Gordon Gibb, Evgenij Belikov, Steven Wei Der Chien, Stefano
Markidis, Pierre Guillou, **Julien Tierny**, Jules Vidal, Charles Gueunet, Johannes
Günther, Mirosław Pawłowski, Piero Poletti, Giorgio Guzzetta, Mattia Manica, Ag-
nese Zardini, Jean-Pierre Chaboureaud, Miguel Mendes, Adrián Cardil, Santiago
Monedero, Joaquín Ramírez, Andreas Gerndt
“*VESTEC: Visual Exploration and Sampling Toolkit for Extreme Computing*”.
IEEE Access
Accepted, 2023.
- ★ Florian Wetzels, Mathieu Pont, **Julien Tierny**, Christoph Garth
“*Merge Tree Geodesics and Barycenters with Path Mappings*”.
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2023.
Best Paper Honorable Mention Award.
- ★ Robin Maack, Jonas Lukasczyk, **Julien Tierny**, Hans Hagen, Ross Maciejewski,
Christoph Garth
“*Parallel Computation of Piecewise Linear Morse-Smale Segmentations*”.
IEEE Transactions on Visualization and Computer Graphics
Accepted, 2023.
- ★ Pierre Guillou, Jules Vidal, **Julien Tierny**,
“*Discrete Morse Sandwich: Fast Computation of Persistence Diagrams for Scalar Data –
An Algorithm and A Benchmark*”.
IEEE Transactions on Visualization and Computer Graphics
Accepted, to be presented at IEEE VIS 2023.
- ★ Malgorzata Olejniczak **Julien Tierny**,
“*Topological Data Analysis of Vortices in the Magnetically-Induced Current Density in
LiH Molecule*”.
Physical Chemistry Chemical Physics
Accepted, 2023.

RESEARCH (continued)

- ★ Mathieu Pont, Jules Vidal, **Julien Tierny**,
“Principal Geodesic Analysis of Merge Trees (and Persistence Diagrams)”.
IEEE Transactions on Visualization and Computer Graphics
Accepted, to be presented at IEEE VIS 2023.
- ★ Mathieu Pont, Jules Vidal, Julie Delon, **Julien Tierny**,
“Wasserstein Distances, Geodesics and Barycenters of Merge Trees”.
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2021.
- ★ Jules Vidal, Pierre Guillou, **Julien Tierny**,
“A Progressive Approach to Scalar Field Topology”,
IEEE Transactions on Visualization and Computer Graphics
To be presented at IEEE VIS 2021.
- ★ Harish Doraiswamy, **Julien Tierny**, Paulo J. S. Silva, Luis Gustavo Nonato, Claudio Silva
“TopoMap: A 0-dimensional Homology Preserving Projection of High-Dimensional Data”,
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2020.
- ★ Jonas Lukasczyk, Christoph Garth, Ross Maciejewski, **Julien Tierny**,
“Localized Topological Simplification of Scalar Data”,
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2020.
- ★ Malgorzata Olejniczak, André Severo Pereira Gomes, **Julien Tierny**,
“A Topological Data Analysis Perspective on Non-Covalent Interactions in Relativistic Calculations”,
International Journal of Quantum Chemistry
Accepted, 2019.
- ★ Jules Vidal, Joseph Budin, **Julien Tierny**,
“Progressive Wasserstein Barycenters of Persistence Diagrams”,
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2019.
Best Paper Honorable Mention Award.
- ★ Charles Gueunet, Pierre Fortin, Julien Jomier, **Julien Tierny**,
“Task-based Augmented Contour Trees with Fibonacci Heaps”,
IEEE Transactions on Parallel and Distributed Systems
Accepted, 2019.
- ★ Guillaume Favelier, Noura Faraj, Brian Summa, **Julien Tierny**,
“Persistence Atlas for Critical Point Variability in Ensembles”,
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2018.
- ★ **Julien Tierny**, Guillaume Favelier, Joshua Levine, Charles Gueunet, Michael Michaux,
“The Topology ToolKit”,
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2017.
Best Paper Honorable Mention Award.
- ★ Brian Summa, **Julien Tierny**, Valerio Pascucci,
“Visualizing the Uncertainty of Graph-based 2D Segmentation with Min-path Stability”,
Computer Graphics Forum
Proc. of EuroVis 2017.
- ★ Ana Maria Vintescu, Florent Dupont, Guillaume Lavoué, Pooran Memari, **Julien Tierny**,
“Least Squares Affine Transitions for Global Parameterization”,
Journal of WSCG
Accepted, 2017.
- ★ Jonas Lukasczyk, Garrett Aldrich, Michael Steptoe, Guillaume Favelier, Charles Gueunet, **Julien Tierny**, Ross Maciejewski, Bernd Hamann, and Heike Leitte,
“Viscous Fingering: A Topological Visual Analytic Approach”,
Applied Mechanics and Materials
Accepted, 2017.

RESEARCH (continued)

- ★ **Julien Tierny** and Hamish Carr
"Jacobi Fiber Surfaces for Bivariate Reeb Space Computation",
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2016.
Best Paper Award.
- ★ Pavol Klačanský, **Julien Tierny**, Hamish Carr, Zhao Geng,
"Fast and Exact Fiber Surfaces for Tetrahedral Meshes",
IEEE Transactions on Visualization and Computer Graphics
Presented at IEEE VIS 2016.
- ★ Roberto A. Boto, Julia C. Garcia, **Julien Tierny**, Jean-Philip Piquemal,
"Interpretation of the reduced density gradient",
Molecular Physics
2016, accepted.
- ★ Hamish Carr, Zhao Geng, **Julien Tierny**, Amit Chattopadhyay, Aaron Knoll,
"Fiber Surfaces: Generalizing Isosurfaces to Bivariate Data",
Computer Graphics Forum
Proc. of EuroVis 2015, accepted.
- ★ Sujin Philip, Brian Summa, **Julien Tierny**, Peer-Timo Bremer, Valerio Pascucci,
"Distributed Seams for Gigapixel Panoramas",
IEEE Transactions on Visualization and Computer Graphics
Accepted, 2014.
- ★ Attila Gyulassy, David Guenther, Joshua Levine, **Julien Tierny**, Valerio Pascucci,
"Conforming Morse-Smale Complexes",
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2014.
- ★ David Guenther, Roberto Álvarez Boto, Julia Contreras Garcia, Jean-Philip Piquemal, **Julien Tierny**,
"Characterizing Molecular Interactions in Chemical Systems",
IEEE Transactions on Visualization and Computer Graphics
Proc. of IEEE VIS 2014.
- ★ David Guenther, Joseph Salmon, **Julien Tierny**,
"Mandatory Critical Points of 2D Uncertain Scalar Fields",
Computer Graphics Forum,
Proc. of EuroVis 2014.
- ★ Fang Chen, Harald Obermaier, Hans Hagen, Bernd Hamann, **Julien Tierny** and Valerio Pascucci,
"Topology analysis of time-dependent multi-fluid data using the Reeb graph",
Computer Aided Geometric Design, 2013.
- ★ Jean-Marc Thiery, **Julien Tierny**, and Tamy Boubekeur,
"Jacobians and Hessians of Mean Value Coordinates for Closed Triangular Meshes",
The Visual Computer Journal, Ed. Springer,
Accepted, 2013.
- ★ **Julien Tierny** and Valerio Pascucci,
"Generalized Topological Simplification of Scalar Fields on Surfaces",
IEEE Transactions on Visualization and Computer Graphics,
Proceedings of IEEE VIS 2012.
- ★ Brian Summa, **Julien Tierny** and Valerio Pascucci,
"Panorama Weaving: Fast and Flexible Seam Processing",
ACM Transactions on Graphics,
Proceedings of ACM SIGGRAPH 2012.
- ★ Jean-Marc Thiery, **Julien Tierny** and Tamy Boubekeur,
"CageR: Cage-based Reverse Engineering of Animated 3D Shapes",
Computer Graphics Forum,
Accepted 2012, Presented at Eurographics 2013.
- ★ Jean-Marc Thiery, Bert Buchholz, **Julien Tierny** and Tamy Boubekeur,
"Analytic Curve Skeletons for 3D Surface Modeling and Processing",
Computer Graphics Forum,
Proceedings of Pacific Graphics 2012.
- ★ **Julien Tierny**, Joel Daniels II, Luis Gustavo Nonato, Valerio Pascucci and Claudio Silva,
"Inspired Quadrangulation",
Computer Aided Design, Ed. Elsevier,
Proceedings of ACM SPM 2011.

RESEARCH (continued)

- * **Julien Tierny**, Joel Daniels II, Luis Gustavo Nonato, Valerio Pascucci and Claudio Silva,
"Interactive Quadrangulation with Reeb Atlases and Connectivity Textures",
IEEE Transactions on Visualization and Computer Graphics,
Accepted, 2011.
- * Tiago Etienne, Luis Gustavo Nonato, Carlos Scheidegger, **Julien Tierny**, Tom Peters, Valerio Pascucci, Mike Kirby and Claudio Silva,
"Topology Verification for Isosurface Extraction",
IEEE Transactions on Visualization and Computer Graphics,
Accepted 2011, Presented at IEEE VIS 2011.
- * Peer-Timo Bremer, Gunther Weber, **Julien Tierny**, Valerio Pascucci, Marc Day and John Bell,
"Interactive Exploration and Analysis of Large Scale Simulations Using Topology-based Data Segmentation",
IEEE Transactions on Visualization and Computer Graphics,
Accepted, 2010.
- * **Julien Tierny**, Attila Gyulassy, Eddie Simon and Valerio Pascucci,
"Loop surgery for volumetric meshes: Reeb graphs reduced to contour trees",
IEEE Transactions on Visualization and Computer Graphics,
Proceedings of IEEE Visualization 2009.
Volume 15, Number 6, 2009, pp. 1177-1184.
- * **Julien Tierny**, Jean-Philippe Vandeboorde and Mohamed Daoudi,
"Partial 3D shape retrieval by Reeb pattern unfolding",
Computer Graphics Forum (Eurographics Association), Ed. Blackwell,
Volume 28, Number 1, 2009, pp. 41-55.
- * **Julien Tierny**, Jean-Philippe Vandeboorde and Mohamed Daoudi,
"Enhancing 3D mesh topological skeletons with discrete contour constrictions",
The Visual Computer Journal Ed. Springer,
Volume 24, Number 3, 2008, pp. 155-172.
- * Sébastien Tchitchek, Mohamed Kissi, **Julien Tierny**,
"Continuous Edit Distance between Time-varying Persistence Diagrams"
IEEE Workshop on Topological Data Analysis and Visualization (TopoInVis) 2026.
- * Mohamed Kissi, Keanu Sisouk, Joshua A. Levine, **Julien Tierny**,
"Topology Aware Neural Interpolation of Scalar Fields"
IEEE Workshop on Topological Data Analysis and Visualization (TopoInVis) 2025.
- * Michael Will, Jonas Lukasczyk, **Julien Tierny**, and Christoph Garth,
"Distributed Path Compression for Piecewise Linear Morse-Smale Segmentations and Connected Components"
IEEE Symposium on Large Data Analysis and Visualization 2024.
- * Fabien Vivodtzev, Florent Nauleau, Jean-Philippe Braeuning, **Julien Tierny**
"Identifying Locally Turbulent Vortices within Instabilities",
IEEE Symposium on Large Data Analysis and Visualization 2024 (posters).
- * Florent Nauleau, Fabien Vivodtzev, Thibault Bridel-Bertomeu, Heloise Beaugendre, and **Julien Tierny**,
"Topological Analysis of Ensembles of Hydrodynamic Turbulent Flows – An Experimental Study"
IEEE Symposium on Large Data Analysis and Visualization 2022.
- * Jules Vidal and **Julien Tierny**,
"Fast Approximation of Persistence Diagrams with Guarantees",
IEEE Symposium on Large Data Analysis and Visualization 2021.
- * Maxime Soler, Martin Petitfrere, Gilles Darche, Melanie Plainchault, Bruno Conche, **Julien Tierny**,
"Ranking Viscous Finger Simulations to an Acquired Ground Truth with Topology-Aware Matchings",
IEEE Symposium on Large Data Analysis and Visualization 2019.
Best Paper Award.
- * Thibault Bridel-Bertomeu, Benjamin Fovet, **Julien Tierny**, Fabien Vivodtzev
"Topological Analysis of High Velocity Turbulent Flow",
IEEE Symposium on Large Data Analysis and Visualization 2019 (posters).

Conferences

RESEARCH (continued)

- ★ Charles Gueunet, Pierre Fortin, Julien Jomier, **Julien Tierny**,
"Task-based Augmented Reeb Graphs with Dynamic ST-Trees",
Eurographics Symposium on Parallel Graphics and Visualization 2019.
- ★ Talha Bin Masood, Joseph Budin, Martin Falk, Guillaume Favelier, Christoph Garth, Charles Gueunet, Pierre Guillou, Lutz Hofmann, Petar Hristov, Adhitya Kamakshidasan, Christopher Kappe, Pavol Klacansky, Patrick Laurin, Joshua A. Levine, Jonas Lukasczyk, Daisuke Sakurai, Maxime Soler, Peter Steneteg, **Julien Tierny**, Will Usher, Jules Vidal, Michal Wozniak
"An Overview of the Topology ToolKit",
Proc. of TopoInVis 2019
- ★ Jonas Lukasczyk, Jakob Beran, Wito Engelke, Martin Falk, Anke Friederici, Christoph Garth, Lutz Hofmann, Ingrid Hotz, Petar Hristov, Wiebke Köpp, Talha Bin Masood, Malgorzata Olejniczak, Paul Rosen, Jan-Tobias Sohns, Tino Weinkauff, Kilian Werner, **Julien Tierny**
"Report of the TopoInVis TTK Hackathon: Experiences, Lessons Learned, and Perspectives",
Proc. of TopoInVis 2019
- ★ Max Kontak, Jules Vidal, **Julien Tierny**,
"Statistical Parameter Selection for Clustering Persistence Diagrams",
Proc. of Super Computing workshop on Urgent HPC 2019
- ★ Maxime Soler, Mélanie Plainchault, Bruno Conche, **Julien Tierny**,
"Lifted Wasserstein Matcher for Fast and Robust Topology Tracking",
IEEE Symposium on Large Data Analysis and Visualization 2018.
Best Paper Honorable Mention Award.
- ★ Maxime Soler, Mélanie Plainchault, Bruno Conche, **Julien Tierny**,
"Topologically Controlled Lossy Compression",
IEEE Pacific Conference on Visualization 2018.
- ★ Charles Gueunet, Pierre Fortin, Julien Jomier, **Julien Tierny**,
Task-based Augmented Merge Trees with Fibonacci Heaps,
IEEE Symposium on Large Data Analysis and Visualization 2017.
- ★ Ana Maria Vintescu, Florent Dupont, Guillaume Lavoué, Pooran Memari, **Julien Tierny**,
"Conformal Factor Persistence for Fast Hierarchical Cone Extraction",
Proc. of Eurographics 2017 (short papers).
- ★ Charles Gueunet, Pierre Fortin, Julien Jomier, **Julien Tierny**.
"Contour Forests: Fast Multi-threaded Augmented Contour Trees",
IEEE Symposium on Large Data Analysis and Visualization 2016.
- ★ Sujin Philip, Brian Summa, **Julien Tierny**, Peer-Timo Bremer, Valerio Pascucci.
"Scalable Seams for Gigapixel Panoramas",
Eurographics Symposium on Parallel Graphics and Visualization 2013.
Best Paper Award.
- ★ Mariem Gargouri, **Julien Tierny**, Erwan Jolivet, Philippe Petit, Elsa Angelini.
"Accurate and robust shape descriptors for the identification of rib cage structures in CT-images with Random Forests",
IEEE International Symposium on Biomedical Imaging 2013.
- ★ Jean-Christophe Michelin, **Julien Tierny**, Florence Tupin, Clément Mallet, and Nicolas Paparoditis,
"Quality Evaluation of 3D City Building Models with Automatic Error Diagnosis",
Proc. of ISPRS Conference on SSG 2013.
- ★ Emanuele Santos, **Julien Tierny**, Ayla Khan, Brad Grimm, Lauro Lins, Juliana Freire, Valerio Pascucci, Claudio Silva, Scott Klasky, Roselyne Barreto, Norbert Podhorszki.
"Enabling Advanced Visualization Tools in a Web-Based Simulation Monitoring System",
IEEE International Conference on eScience 2009.
- ★ Peer-Timo Bremer, Gunther Weber, **Julien Tierny**, Valerio Pascucci, Marcus Day, John Bell.
"A Topological Framework for the Interactive Exploration of Large Scale Turbulent Combustion",
IEEE International Conference on eScience 2009.
- ★ **Julien Tierny**, Jean-Philippe Vandeborre and Mohamed Daoudi,
"Fast and precise kinematic skeleton extraction of 3D dynamic meshes",
IEEE International Conference on Pattern Recognition 2008, pp. 1-4.

RESEARCH (continued)

- * **Julien Tierny**, Jean-Philippe Vandeborre and Mohamed Daoudi,
"Reeb chart unfolding based 3D shape signatures",
Eurographics 2007, short paper, pp. 13-16.
- * **Julien Tierny**, Jean-Philippe Vandeborre and Mohamed Daoudi,
"Topology driven 3D mesh hierarchical segmentation",
IEEE Shape Modeling International 2007, short paper, pp. 215-220.
- * **Julien Tierny**, Jean-Philippe Vandeborre and Mohamed Daoudi,
"3D mesh skeleton extraction using topological and geometrical analyses",
Pacific Graphics 2006, pp. 85-94.
- * **Julien Tierny**, Jean-Philippe Vandeborre and Mohamed Daoudi,
"Invariant high level Reeb graphs of 3D polygonal meshes",
IEEE 3DPVT 2006, pp. 105-112.
- Book chapters
 - * **Julien Tierny**, David Guenther, and Valerio Pascucci,
"Optimal General Simplification of Scalar Fields on Surfaces",
Chapter of *"Topological and Statistical Methods for Complex Data"*,
Springer, 2014 (ISBN: 978-3-662-44899-1).
 - * Stefano Berretti, Mohamed Daoudi, Alberto Del Bimbo, Tarik Filali Ansary, Pietro Pala, **Julien Tierny** and Jean-Philippe Vandeborre,
"3D object indexing", chapter of *"3D object processing: compression, indexing and watermarking"*,
Wiley, June 2008 (ISBN: 978-0-470-06542-6).
- Invited Conferences
 - * **Julien Tierny**, Jean-Philippe Vandeborre and Mohamed Daoudi,
"Geometry flavored topological skeletons: applications to shape handling, understanding and retrieval",
Second DELOS Conference, 2007.
 - * Mohamed Daoudi, Tarik Filali-Ansary, **Julien Tierny** and Jean-Philippe Vandeborre,
"3D mesh models: view-based indexing and structural analysis",
First DELOS Conference, 2007, Lecture Notes in Computer Science, pp. 298-307.
- Technical Reports
 - * Christoph Garth, Charles Gueunet, Pierre Guillou, Federico Iuricich, Joshua A Levine, Jonas Lukasczyk, Mathieu Pont, **Julien Tierny**, Jules Vidal, Bei Wang, Florian Wetzels
"Topological Analysis of Ensemble Scalar Data with TTK, A Sequel".
IEEE VIS Tutorials 2022.
 - * Christoph Garth, Charles Gueunet, Pierre Guillou, Lutz Hofmann, Joshua A Levine, Jonas Lukasczyk, **Julien Tierny**, Jules Vidal, Bei Wang, Florian Wetzels
"Topological Analysis of Ensemble Scalar Data with TTK".
IEEE VIS Tutorials 2021.
 - * Martin Falk, Christoph Garth, Charles Gueunet, Pierre Guillou, Attila Gyulassy, Lutz Hofmann, Christopher Kappe, Joshua A Levine, Jonas Lukasczyk, **Julien Tierny**, Jules Vidal,
"Topological Data Analysis Made Easy with the Topology ToolKit, What is New?".
IEEE VIS Tutorials 2020.
 - * Martin Falk, Christoph Garth, Charles Gueunet, Joshua A Levine, Jonas Lukasczyk, **Julien Tierny**, Jules Vidal,
"Topological Data Analysis Made Easy with the Topology ToolKit, A Sequel".
IEEE VIS Tutorials 2019.
 - * Guillaume Favelier, Charles Gueunet, Attila Gyulassy, Julien Jomier, Joshua Levine, Jonas Lukasczyk, Daisuke Sakurai, Maxime Soler, **Julien Tierny**, Will Usher, Qi Wu
"Topological Data Analysis Made Easy with the Topology ToolKit".
IEEE VIS Tutorials 2018.
 - * Guillaume Favelier, Charles Gueunet, **Julien Tierny**
"Visualizing Ensembles of Viscous Fingers".
IEEE Scientific Visualization Contest 2016.
Honorable Mention Award.
Poster presentation at IEEE VIS 2016.
 - * Brian Summa, **Julien Tierny**, Peer-Timo Bremer, Giorgio Scorzelli, and Valerio Pascucci,
"Active-Stitching: Beyond Batch Processing of Panoramas", 2013.
- Patents**
 - * *"Seam Network Processing for Panorama Weaving"*
Brian Summa, Valerio Pascucci, **Julien Tierny**
US Patent 20,140,002,488.

Distributed Software

★ *the Topology ToolKit*

Open-source C++ library and software collection for efficient and generic topological data analysis and visualization (BSD license).

<http://topology-tool-kit.github.io>

★ *Reeb Space*

Open source library (AGPL license) for bivariate Reeb space computation (implementation of the paper: “Jacobi Fiber Surfaces for Bivariate Reeb Space Computation”).

★ *Fiber Surfaces*

Open source library for fiber surface computation (implementation of the paper: “Fast and exact fiber surfaces for tetrahedral meshes”).

★ *Generalized Topological Simplification of Scalar Fields on Surfaces*

Open source library (LGPL license) for general topological simplification on surfaces (implementation of the paper: “Generalized Topological Simplification of Scalar Fields on Surfaces”, proc. of IEEE VIS 2012).

★ *vtkReebGraph*

Deployment of Reeb graph based visualization techniques into the OpenSource project *Visualization ToolKit* (VTK).

★ *SINAMIS Is Not A Mesh Indexing System*,

Benchmarking tools and partial 3D shape retrieval system implementation (paper: “Partial 3D shape retrieval by Reeb pattern unfolding”).

Keynote Speaker

- 2024 ★ IEEE VIS Workshop on Progressive Data Analysis and Visualization, USA.
- 2020 ★ EuroVis VISGAP Symposium, Sweden.
- 2015 ★ IEEE Shape Modeling International, France.

Invited Talks

- 2026 ★ Curves and Surfaces (TDA symposium), France.
- ★ Geometry, topology and machine learning, France.
- ★ Point Processes and their Applications, France.
- 2025 ★ CECAM Workshop computational topology for chemical systems, Switzerland.
- 2022 ★ Workshop on Computational Persistence, USA.
- ★ Workshop on Topological Data Visualization, USA.
- ★ IMSI Workshop on the Mathematics of Soft Matter, USA.
- ★ Workshop on Applied Topology in Frontier Sciences, Singapore.
- 2020 ★ Huawei Research Center, Moscow, Russia.
- ★ Applied Algebraic Topology Network (Webinar).
- 2018 ★ ACM SIGGRAPH, Vancouver, Canada (invited IEEE TVCG papers).
- ★ French conference on computer science and geometry, Lyon, France.
- 2017 ★ ACM SIGGRAPH, Los Angeles, USA (invited IEEE TVCG papers).
- ★ Center for Data Science at New York University, USA.
- ★ Topology, Computation and Data Analysis, Dagstuhl, Germany.
- ★ Conference on Physical Modeling for Virtual Manufacturing Systems and Processes, Germany.
- ★ Imaging in Paris, Institut Henri Poincaré, France.
- ★ GdR MASCOT-NUM workshop, Institut Henri Poincaré, France.
- ★ INRIA Saclay, France.
- ★ Telecom ParisTech, France.
- ★ Strasbourg university, France.
- 2016 ★ Heidelberg University, Germany.
- 2015 ★ Tulane University, USA.
- 2014 ★ ENS Cachan CMLA, France.
- ★ Uncertainty Forum, CEA, France.
- ★ Franco-Romanian Applied Math Congress, Lyon, France.
- ★ GdR-ISIS AC3D Workshop, Porquerolles, France.
- ★ Leeds University, UK.
- ★ LIRIS, Lyon University, France.
- 2013 ★ Ceremade Seminar, Paris Dauphine University, France.
- 2012 ★ Max Planck Institut für Informatik - Saarbrücken, Germany.
- ★ Clemson University, USA.
- 2008 ★ IGG Research Group, LSIT, Strasbourg, France.
- ★ LAIC Laboratory, Clermont Ferrand, France.
- ★ Alice Research Group, INRIA Loria, Nancy, France.

R E S E A R C H (continued)

2006	★ Tours University, Tours, France.
	<i>Professional service</i>
Associate Editor	★ IEEE Transactions on Visualization and Computer Graphics (2018-2022). ★ Computer Graphics Forum (2023 – 2026).
International Steering Committee	★ IEEE LDAV. ★ TopoInVis workshop series (co-chair).
General Chair	★ IEEE LDAV 2020-2021.
Program Chair	★ IEEE VIS Short Papers 2025. ★ IEEE VIS Workshops 2023. ★ TopoInVis 2019, 2022 (full papers). ★ IEEE LDAV 2018-2019 (full papers). ★ EG Symposium on Parallel Graphics & Visualization 2022 (full papers). ★ IEEE LDAV 2014 (poster track).
Program Committee	★ IEEE VIS 2017-2019, 2021-2023, 2025-2026 (full papers). ★ EuroVis 2015-2017, 2020-2022, 2024-2026 (full papers). ★ TopoInVis 2015, 2017 (full papers). ★ EG Symposium on Parallel Graphics and Visualization 2017-2019 (full papers). ★ SuperComputing 2017-2018 (tutorials). ★ IEEE Shape Modeling International 2015 (full papers). ★ EuroVis 2013-2014 (short papers). ★ Eurographics 2012-2013 (short papers). ★ Graphics Replicability Stamp Initiative (2017-).
Journals	★ Reviewer for: IEEE Transactions on Visualization and Computer Graphics, Computer Graphics Forum, Computer-Aided Design, Computer-Aided Geometric Design, IEEE Transactions on Image Processing, International Journal of Computer Vision, Theoretical Computer Science, Image and Vision Computing.
Conferences	★ Reviewer for: ACM SIGGRAPH (2012-2013, 2015, 2017), ACM SIGGRAPH Asia (2015), Eurographics (2009-2012, 2015-2017), Pacific Graphics (2011), ACM Solid and Physical Modeling (2008), IEEE Shape Modeling International (2008, 2015), TopoInVis (2013, 2015, 2017), High Performance Graphics (2013), IEEE SIBGRAPI (2009), IEEE ICME (2007-2008).
Recruitment Committee	★ Assistant professor in mathematics, Sorbonne University, 2025. ★ Assistant professor in applied mathematics or computer science, ENSAM, 2024. ★ Permanent research engineer, INRIA Saclay, 2017. ★ Assistant professor in mathematics, applied mathematics or computer science, Sorbonne Université, 2016.
Ph.D. Thesis Reviewer	★ Guillaume Brouillette (Sherbrooke University), 2025. ★ Davide Gurnari (University of Warsaw), 2025. ★ Charles Lepaire (University of Poitiers), 2025. ★ Erwan Duhamel (Université de Strasbourg), 2024. ★ Alexandre Bois (ENS Paris Saclay), 2024. ★ Welcome Alexandre-Barff (University of Reims), 2024. ★ Maxime Kirgo (École Polytechnique), 2022. ★ Christopher Kappe (TU Kaiserslautern), 2022. ★ Kilian Werner (TU Kaiserslautern), 2022. ★ Nicolas Le Goff (CEA), 2020. ★ Jonas Lukasczyk (TU Kaiserslautern), 2019. ★ Steve Petruzza (University of Utah), 2018. ★ Arnaud Bleterrer (Université Côte d'Azur), 2018.
Ph.D. Thesis Committee	★ François Clément (Sorbonne Université), 2024. ★ Florent Nauleau (Bordeaux University), 2023. ★ Gonzalo Romero-García (IRCAM), 2023. ★ Pavol Klacansky (University of Utah), 2023. ★ Thomas Gerald (Sorbonne Université - Paris), 2020. ★ Alexandre Bone (INRIA - Paris), 2020. ★ Maxime Louis (INRIA - Paris), 2019. ★ Roberto Alvarez-Boto (Sorbonne Université), 2016. ★ Leo Allemand-Giorgis (INRIA - Grenoble), 2016. ★ Esma Elghoul (INRIA - Rocquencourt), 2014. ★ Maxime Belperin (LIRIS), 2013. ★ Rachid El Khoury (LIFL), 2013. ★ Bertrand Pellenard (INRIA - Sophia Antipolis), 2012. ★ Romain Arcila (LIRIS), 2011.

RESEARCH (continued)

National conferences	★ Visu 2020 (co-organizer, virtual event). ★ Visu 2012 (co-organizer). ★ Visu 2011-2015 (program committee member). ★ Coresa 2012 (program committee member).
Funding agencies (reviewer)	★ National Science Foundation (NSF). ★ European Commission (H2020). ★ Fulbright fellowships (Franco-American Commission). ★ Agence Nationale de la Recherche (ANR). ★ Institut Mines-Télécom.
National responsibilities	★ Co-head of the Visualization work-group (“GT Visu”) of the CNRS GdR IGRV (2016-2021).

ACADEMIC ACTIVITIES

Current teaching activities (around 64 hours per year)

ENS Paris Saclay	★ Head instructor for the Topological Data Analysis course (16 hours/year) with Frédéric Chazal, MVA Master (2nd year).
Ecole Polytechnique	★ Introductions to C++ programming (48 hours).

Past teaching activities

2017 – 2025	★ Introductions to Topological Data Analysis (3 hours) and Scientific Visualization (3 hours) at Telecom ParisTech.
2017 – 2024	★ Head instructor for the Visual Data Analysis course at ENS Paris Saclay (MPRI Master, 24 hours/year) with Jean-Daniel Fekete, MPRI Master (2nd year).
2015 – 2021	★ Head instructor for the Visualization course at Sorbonne PolyTech (35 hours/year), Master 2 level.
2013 – 2021	★ Head instructor for the Visualization course at ENSTA (27 hours/year), Master 2 level.
2017	★ Winter school on Computational Geometry and Topology for Data Analysis, Nice, France (organizers: Jean-Daniel Boissonnat and Frédéric Chazal).
2014	★ ICS summer school on Scientific Visualization, Roscoff, France (organizer: Pascal Frey).
2013 – 2017	★ Head instructor for the Visualization course, Versailles University (UVSQ).
2008 – 2010	★ Teaching fellow at the Computer Science Department of the University of Utah.
2005 – 2008	★ Teaching assistant (“Moniteur”) at the Computer Science Department of Lille University.